

The Conley index and the topological dimension of hyperbolic Julia sets

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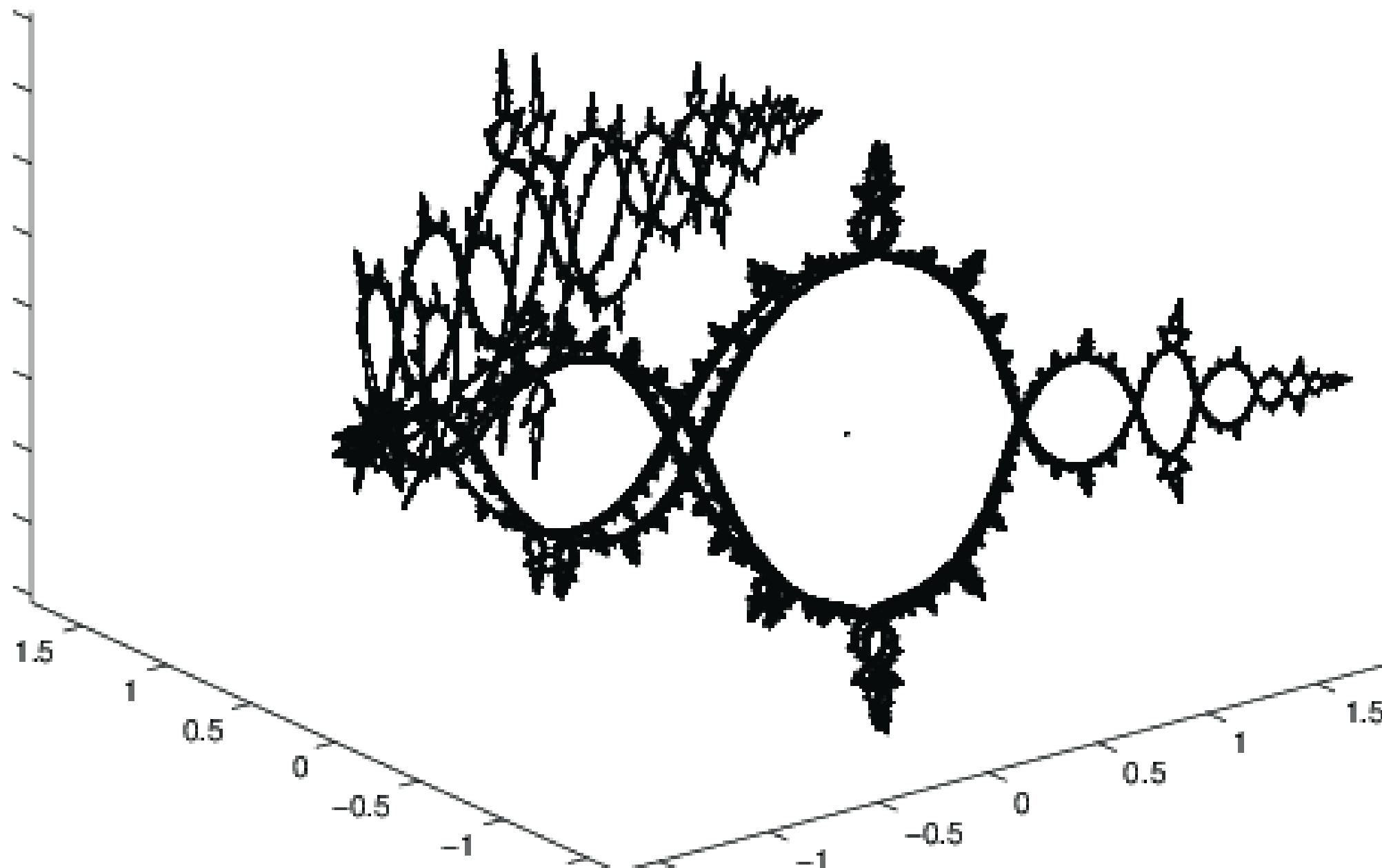


Institute of
SCIENCE TOKYO



Result

Theorem. Assume that the Hénon map has some attracting periodic orbits and is hyperbolic on J . Then the topological dimension of J is 1.



Julia Set for Quadratic Maps

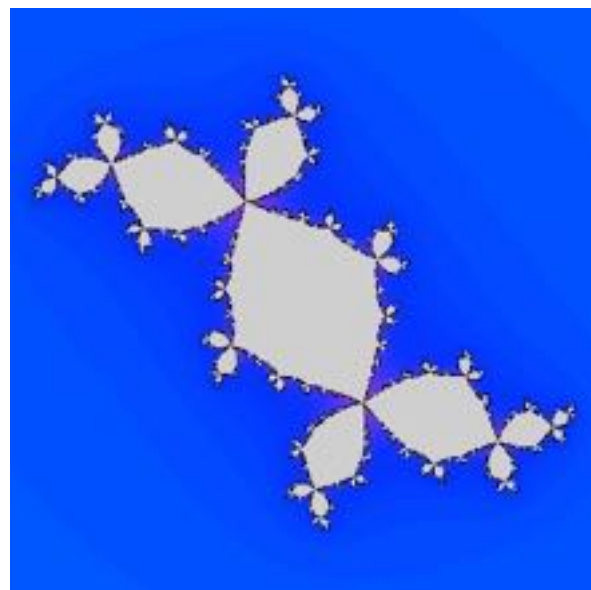
The complex quadratic map

$$P_c : \mathbb{C} \rightarrow \mathbb{C} : z \mapsto z^2 + c \quad (c \in \mathbb{C} \text{ is the parameter})$$

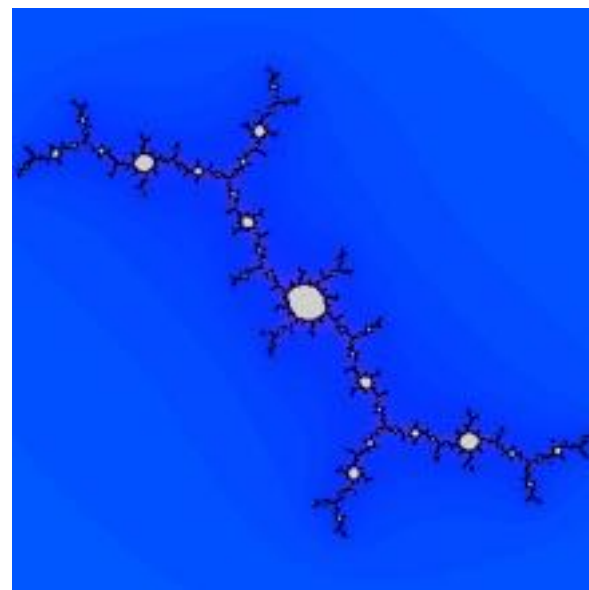
$$P_c^n := P_c \circ P_c \circ \cdots \circ P_c \text{ (n-times)}$$

The filled Julia set and the Julia set

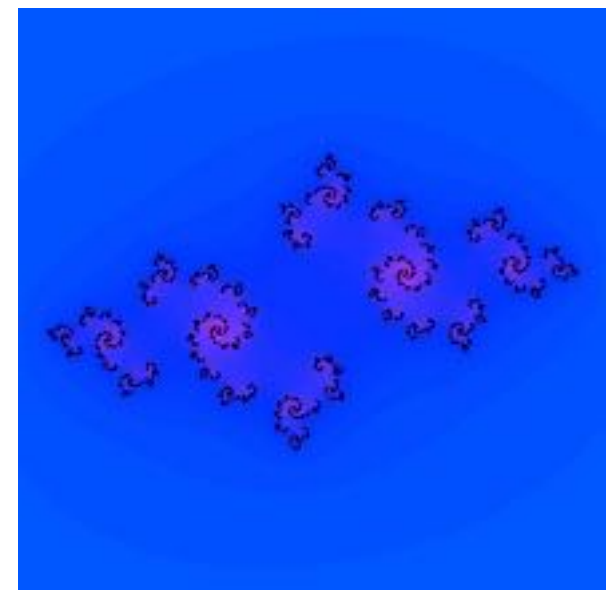
$$K_c := \{z \in \mathbb{C} \mid \{P_c^n(z)\} \text{ is bounded}\} \quad J_c := \partial K_c$$



$$c = -0.125475 + 0.7494i$$



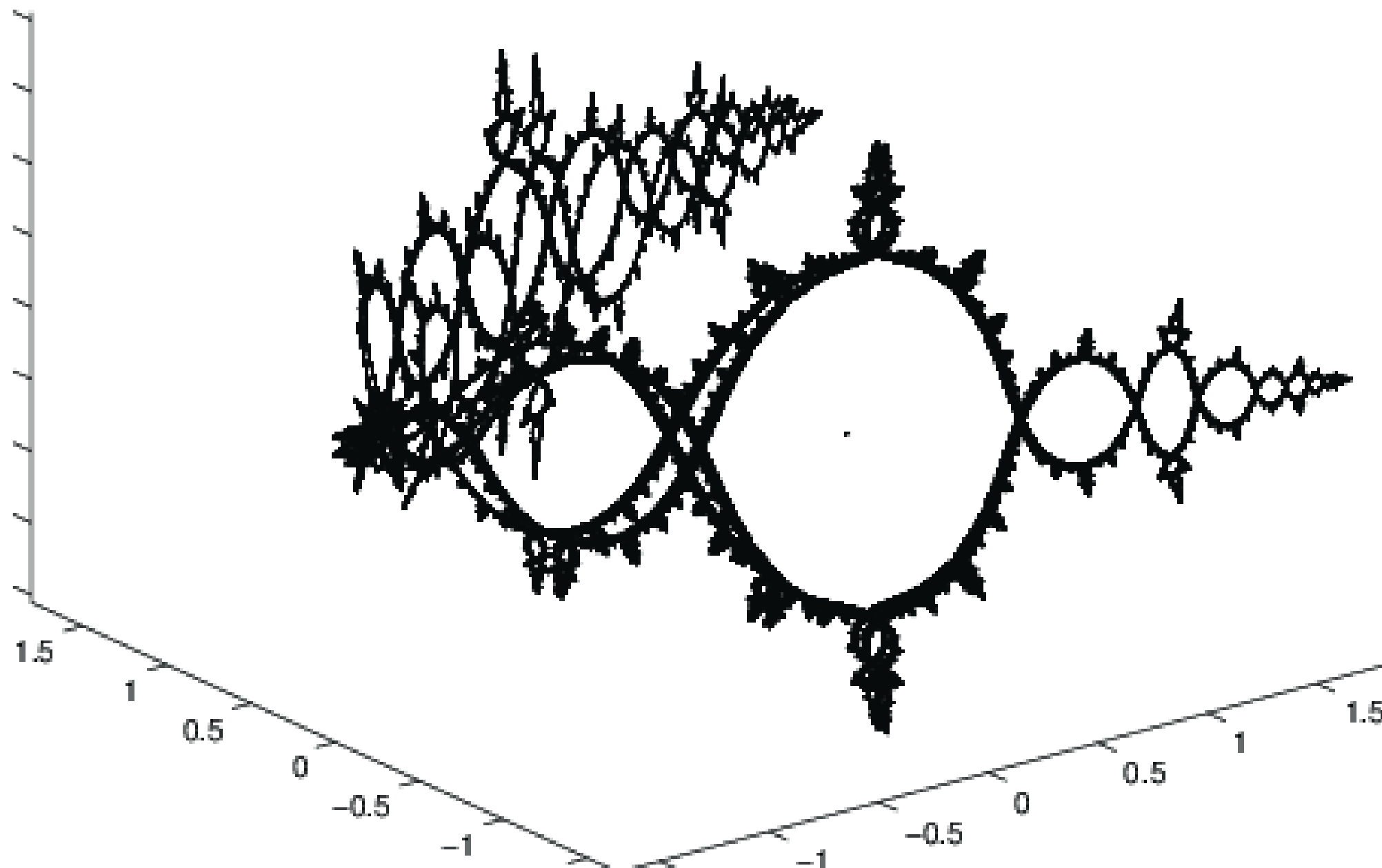
$$c = -0.157414272428 + 1.033054947853i$$



$$c = -0.810000061989 - 0.344999969006i$$

Result

Theorem. Assume that the Hénon map has some attracting periodic orbits and is hyperbolic on J . Then the topological dimension of J is 1.



Conley index

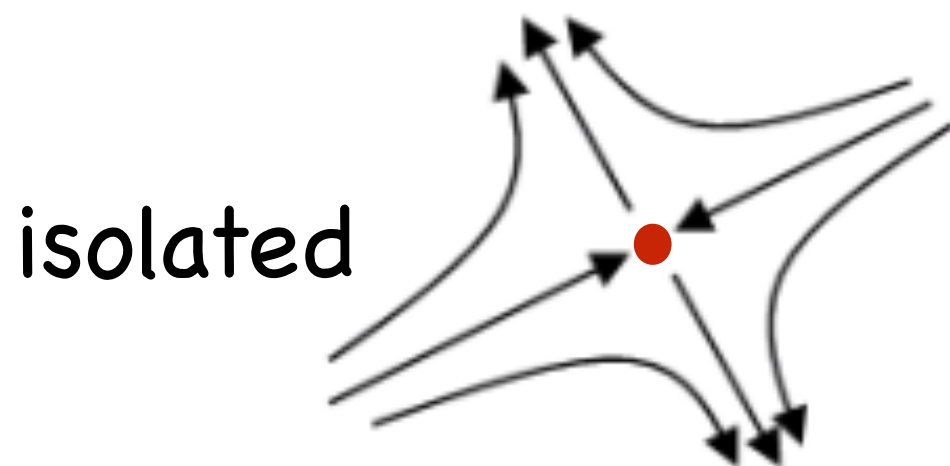
Isolated Invariant Set

$f : X \rightarrow X$: a homeomorphism

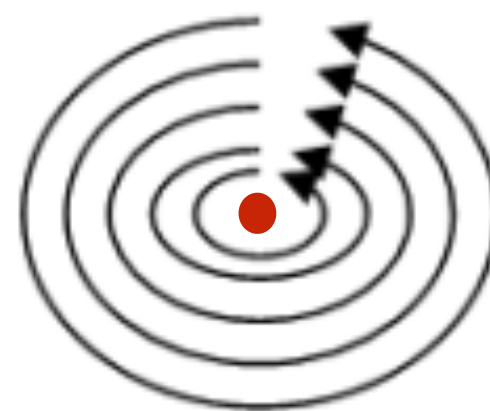
Definition. An invariant set S is isolated if there is a compact neighborhood N of S such that

$$S = \text{Inv}(N, f) := \{x \in X \mid f^n(x) \in N \text{ for all } n \in \mathbb{Z}\}$$

holds. Then N is called an isolating neighborhood.



isolated



not isolated

Remark. Being an isolating nbd is a robust property.

Index Pair

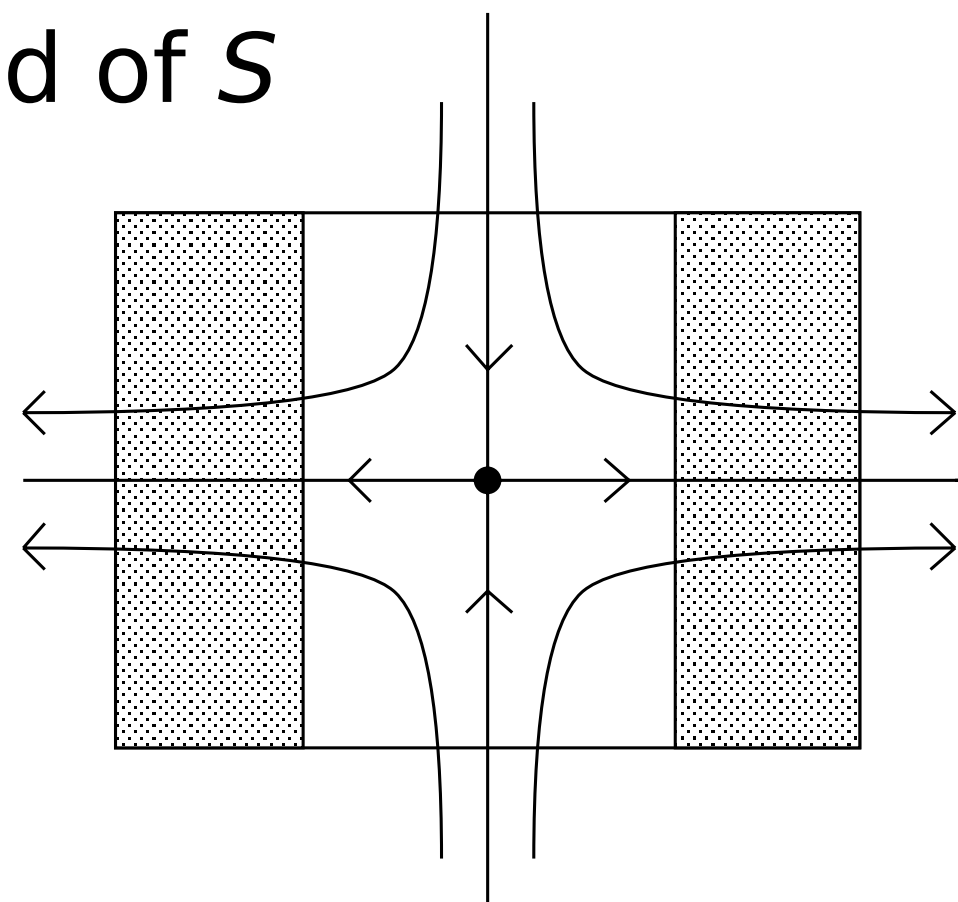
S : an isolated invariant set

Definition. An index pair for S is a pair (N, L) of compact sets $L \subset N$ such that

- * $\text{cl}(N \setminus L)$ is an iso. nbd of S
- * $f(L) \cap N \subset L$
- * $f(N \setminus L) \subset N$

If $P = (N, L)$ is an index pair,
 $f_P : N/L \rightarrow N/L$ defined by

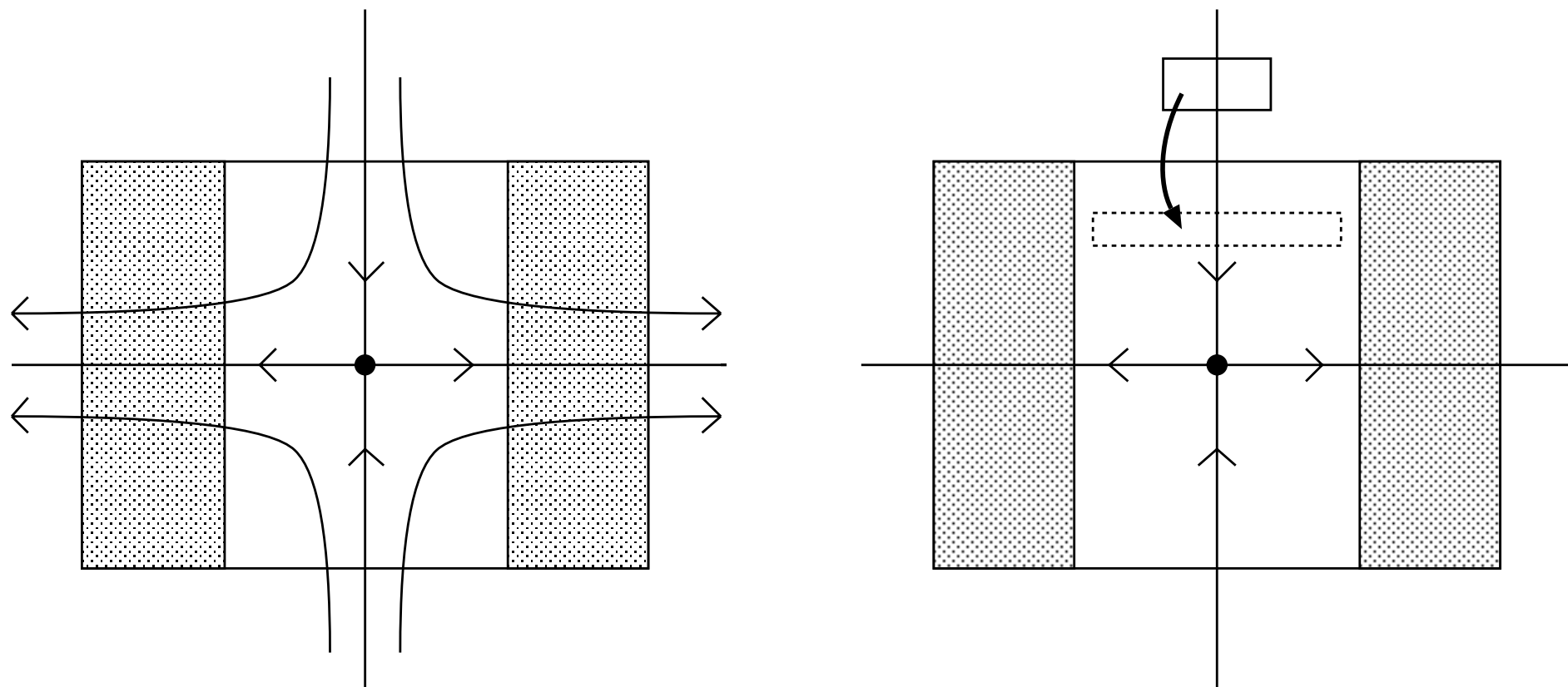
$$f_P([x]) := \begin{cases} [f(x)] & \text{if } f(x) \in N \\ [L] & \text{otherwise} \end{cases}$$



is continuous.

Redundancy

We want to say that the homotopy type or the homology class of $f_P : N/L \rightarrow N/L$ is a topological invariant of S , however, this is not well defined.



Two topologically different index pairs for a saddle.

Shift Equivalence

Definition. We say $f : X \rightarrow X$ and $g : Y \rightarrow Y$ are shift equivalent if there exist a natural number m and $r : X \rightarrow Y$, $s : Y \rightarrow X$ such that

$$r \circ f = g \circ r, \quad s \circ g = f \circ s, \quad r \circ s = g^m, \quad s \circ r = f^m$$

$$\begin{array}{ccc} X & \xrightarrow{f} & X \\ r \downarrow & & \downarrow r \\ Y & \xrightarrow{g} & Y \end{array} \qquad \begin{array}{ccc} X & \xrightarrow{f} & X \\ s \uparrow & & \uparrow s \\ Y & \xrightarrow{g} & Y \end{array}$$

Remark. In the category of integer matrices acting on $\mathbb{Z}^n$, “The Jordan form away from zero” is invariant under the shift equivalence.

The Homology Conley Index

Definition. The homology Conley index of S is the shift equivalence class of

$$f_{P*} : H_*(N/L, [L]) \rightarrow H_*(N/L, [L])$$

where $P = (N, L)$ is an index pair of S .

This is well defined!

Remark. The Conley index is a generalization of the Morse index. In fact, the Conley index of a saddle fixed point with Morse index u is

$$\begin{cases} \pm 1 : \mathbb{Z} \rightarrow \mathbb{Z} & \text{at degree } u \\ 0 & \text{otherwise} \end{cases}$$

(the sign is $+$ iff the map preserve the orientation)

Basic Properties

Ważewski Property

If the Conley index of S is non-trivial, then S is not empty.

Fixed Point Theorem

If the Lefschetz number of f_P* is non-zero, there exists a fixed point in S .

Entropy Estimation

If f is a smooth diffeomorphism, $h_{\text{top}}(f|_S)$ is greater than or equal to the logarithm of the spectral radius of the Conley index.

Continuation Property

Consider a one-parameter family $f_\lambda : X \rightarrow X$, $\lambda \in \mathbb{R}$

Definition. Let N be a compact set. We say that $S_{\lambda_0} = \text{Inv}(N, f_{\lambda_0})$ is a continuation of $S_{\lambda_1} = \text{Inv}(N, f_{\lambda_1})$ if N is isolating for all $\lambda \in [\lambda_0, \lambda_1]$.

(And we extend this relation transitively)

Continuation Property

If S' is a continuation of S , then their Conley indices coincide.

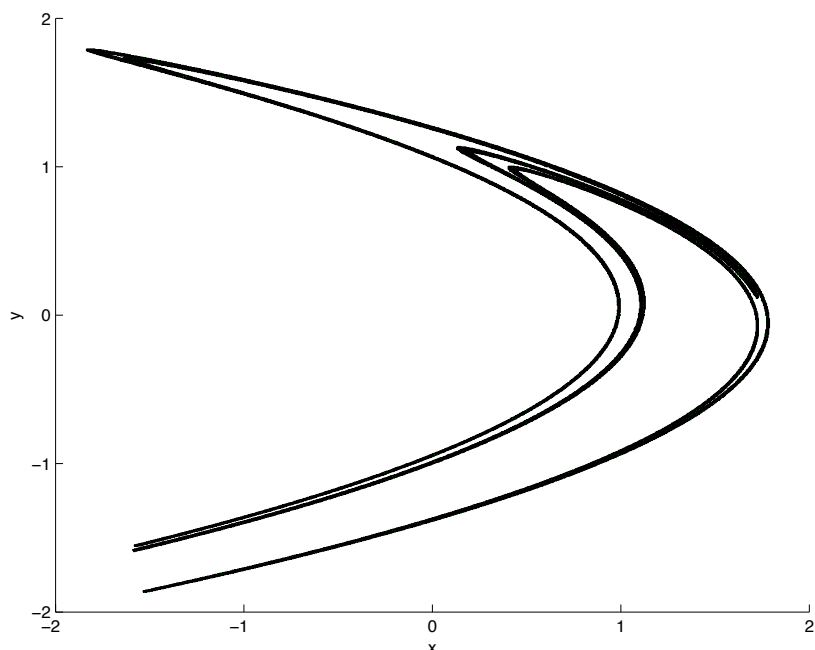
Conley index for the real and complex Hénon map

The Hénon Map

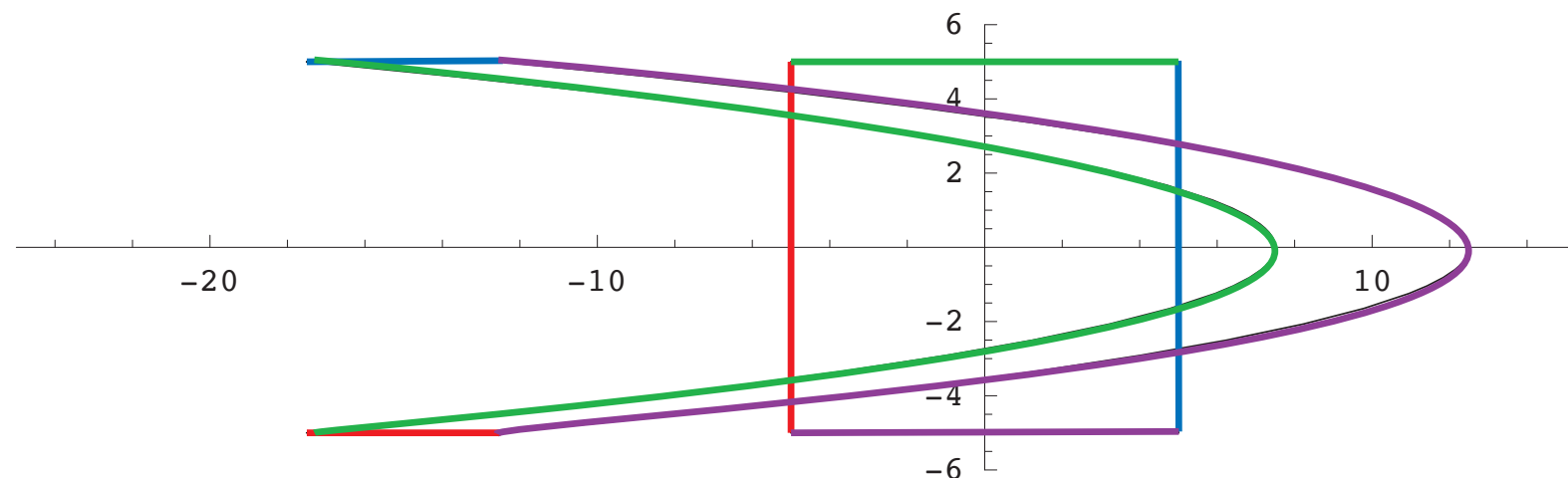
$$H_{a,b} : \mathbb{R}^2 \rightarrow \mathbb{R}^2 \quad (\mathbb{C}^2 \rightarrow \mathbb{C}^2)$$

$$(x, y) \mapsto (a - x^2 + by, x)$$

where a and b are real (complex) parameters.



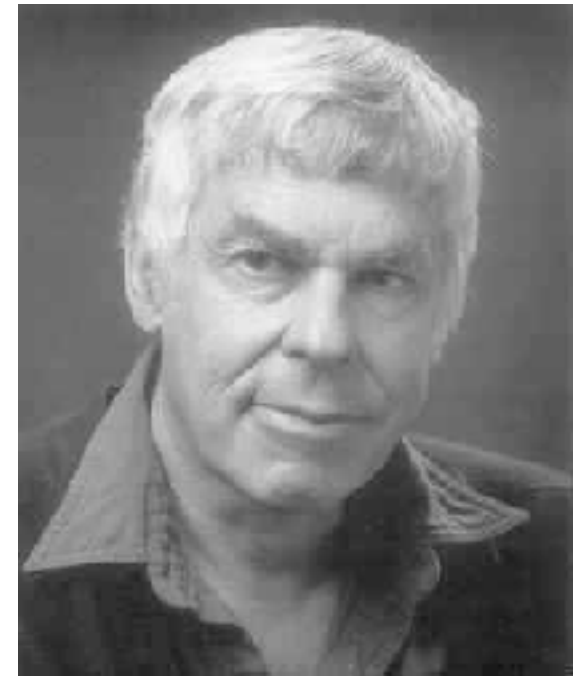
The “strange” attractor at
 $(a, b) = (1.4, 0.3)$



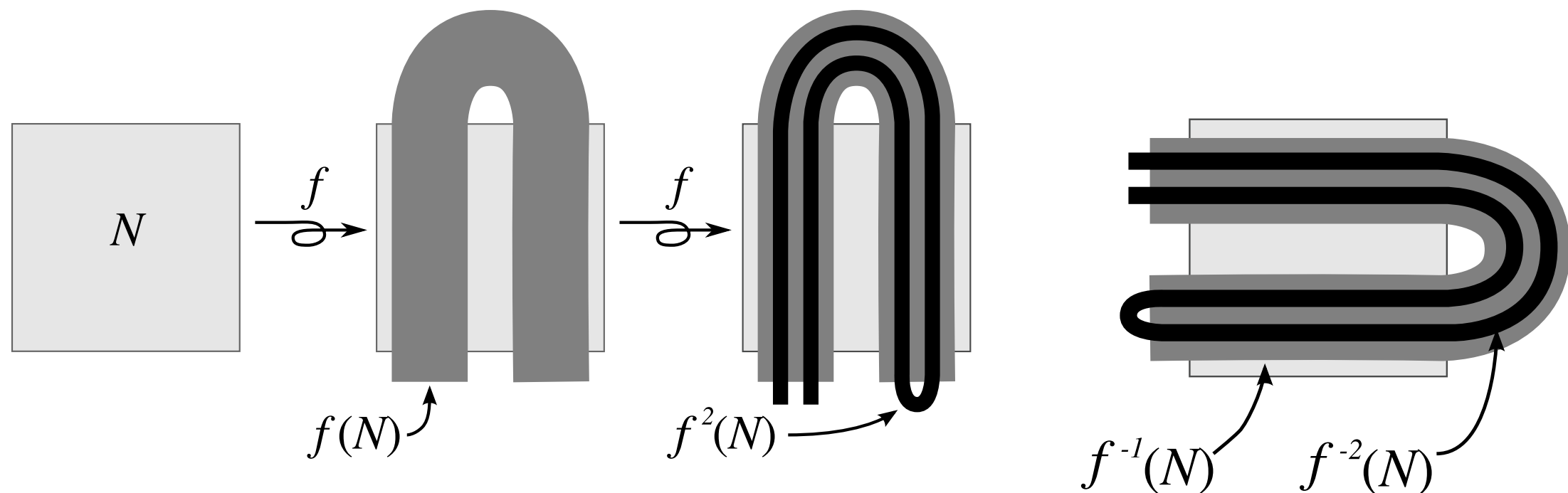
Devaney-Nitecki Horseshoe for large a

[Devaney and Nitecki 1979] For any fixed b , if a is sufficiently large, then the non-wandering set of the Hénon map is uniformly hyperbolic full horseshoe.

A Geometric Model of Chaos



On the beach of Rio,
Smale found a horseshoe...



Index of the real Hénon

$$K^{\pm} := \{p \in \mathbb{C}^2 \mid \{H_{a,b}^{\pm n}(p) \text{ for } n = 0, 1, 2, \dots\} \text{ is bdd}\}$$

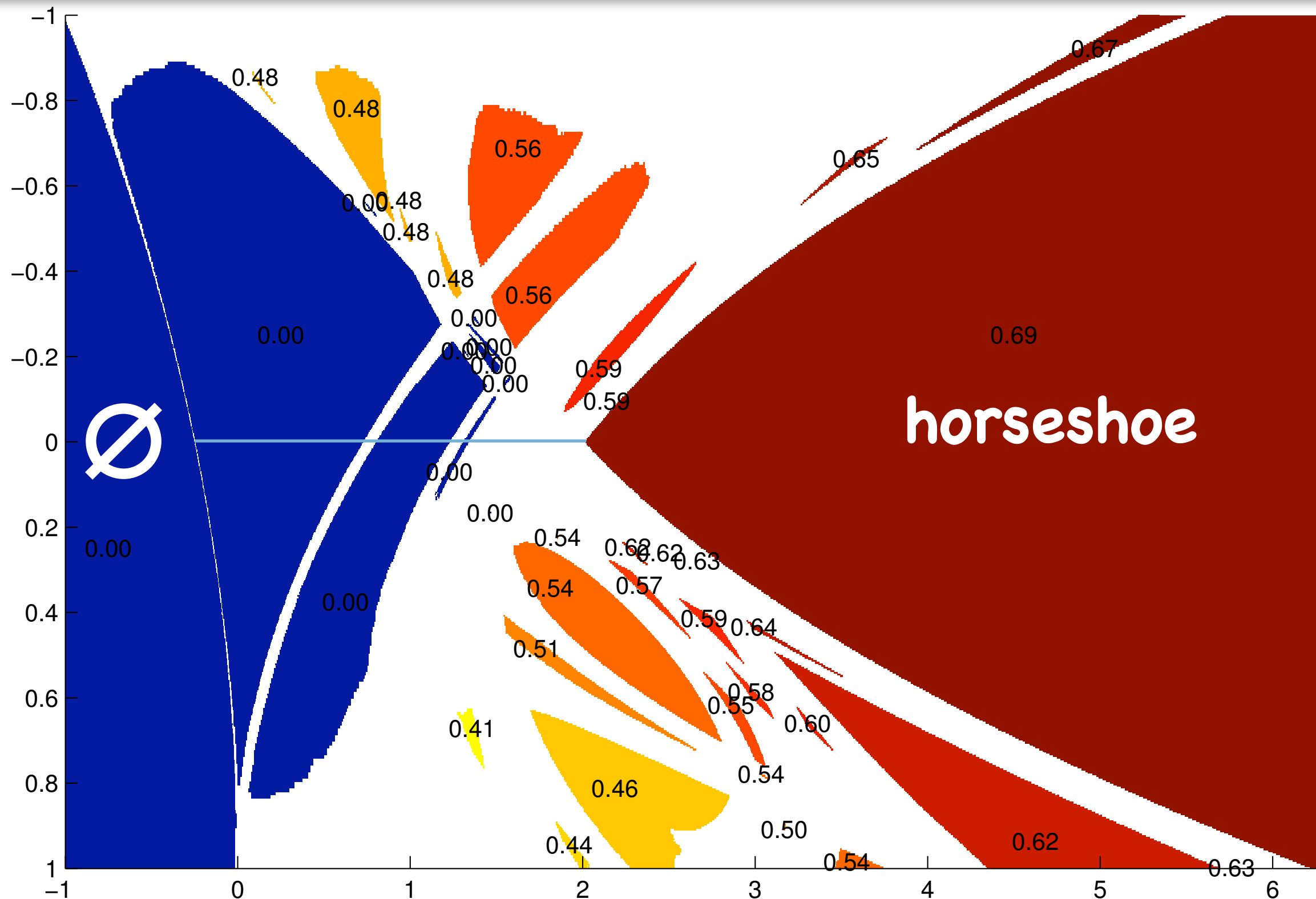
$$K := K^{+} \cap K^{-} \quad J^{\pm} := \partial K^{\pm} \quad J := J^{+} \cap J^{-}$$

$$K^{\mathbb{R}} := K \cap \mathbb{R}^2$$

Theorem. For all $(a, b) \in \mathbb{R}^2$, the Conley index of $K^{\mathbb{R}}$ is trivial (at all dimensions).

Proof. Consider a large rectangle N that isolates $K^{\mathbb{R}}$ for all $-1 \leq a \leq 8$, $-1 \leq b \leq 1$. Since $K^{\mathbb{R}}$ is empty at $(a, b) = (-1, 0)$ and therefore the index is trivial at this parameter value, the theorem follows from the continuation property of the Conley index.

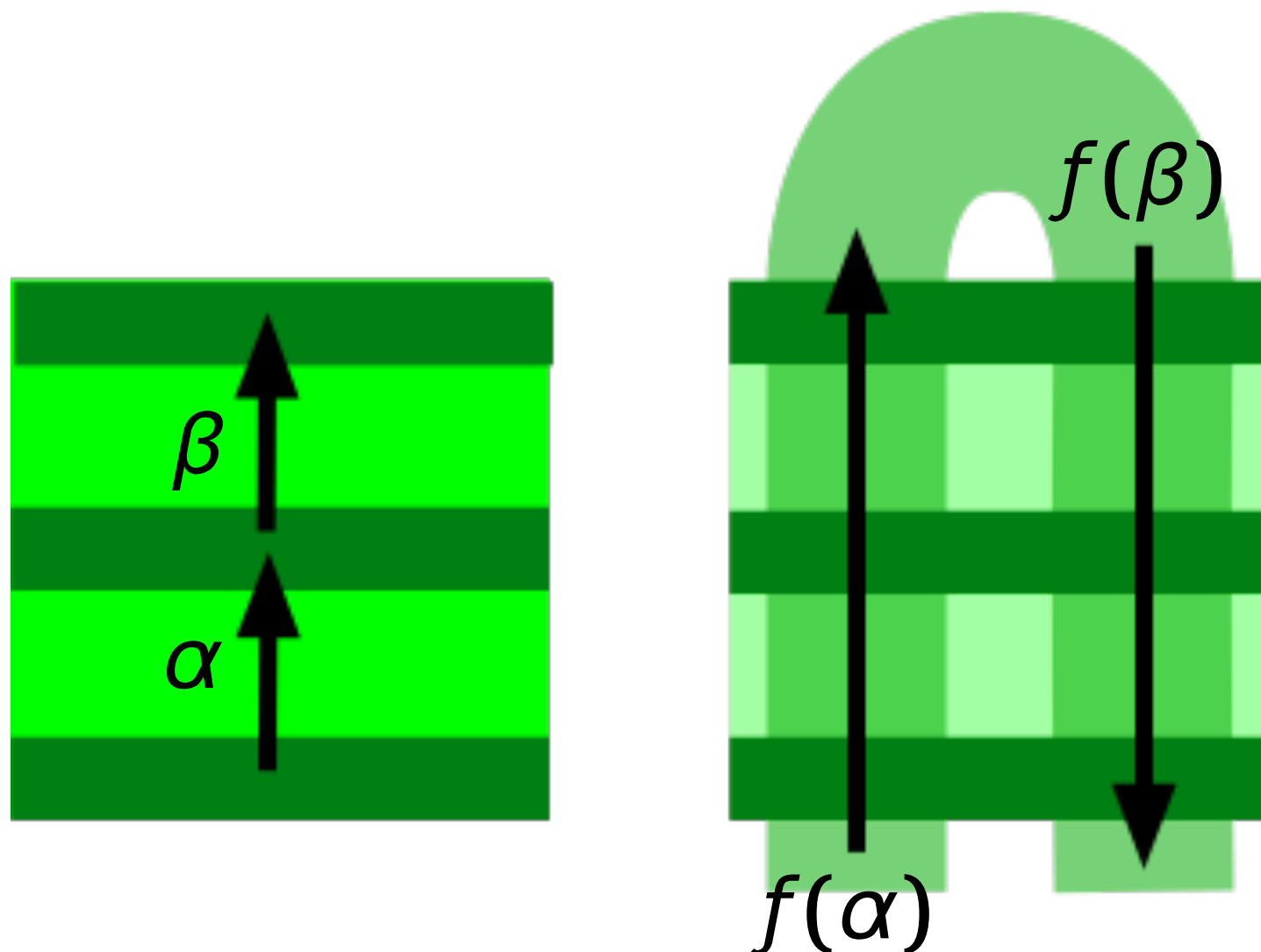
Creation of horseshoe



Hyperbolic plateaus of the real Hénon map $(x, y) \mapsto (a - x^2 + by, x)$
and rigorous lower bounds of the topological entropy

Index of the Horseshoe

Direct computation of the index of the horseshoe:



$$\alpha \mapsto \alpha + \beta$$

$$\beta \mapsto -\alpha - \beta$$

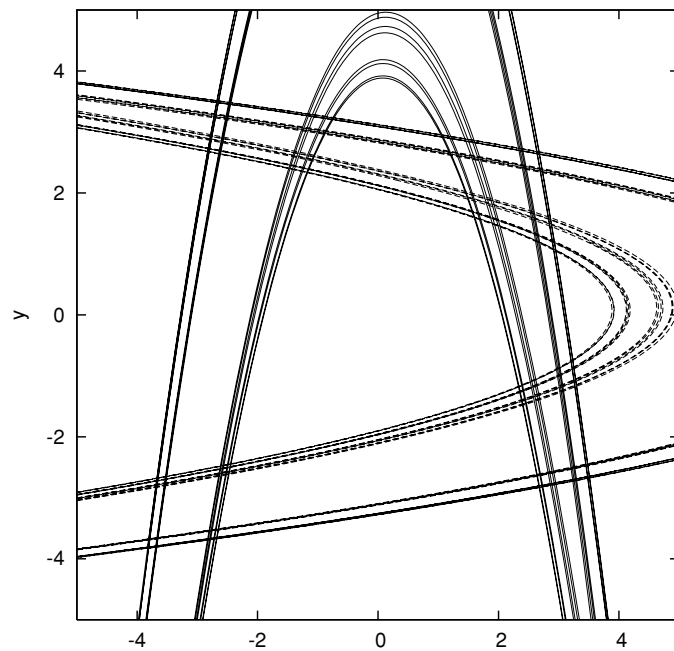
$$\begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} : \mathbb{Z}^2 \rightarrow \mathbb{Z}^2$$

is shift equiv. to 0

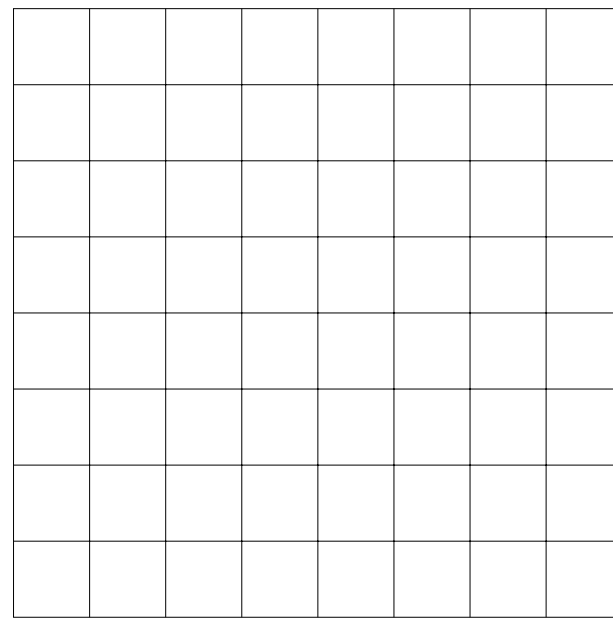
Remark. We can disregard the orientation by using $\mathbb{Z}_2$ coefficient, however, the $\mathbb{Z}_2$ index is also trivial since $2 = 0$ in $\mathbb{Z}_2$.

Pruning Front

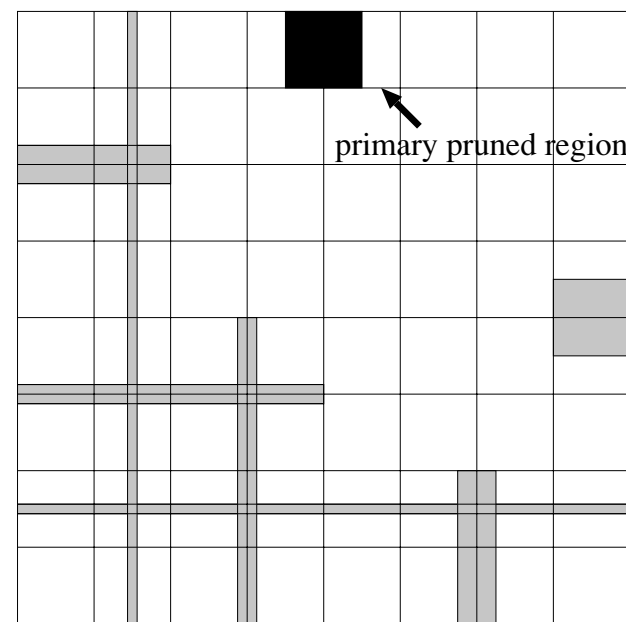
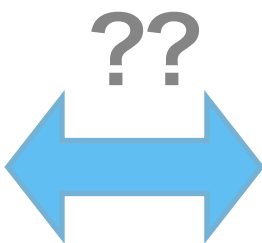
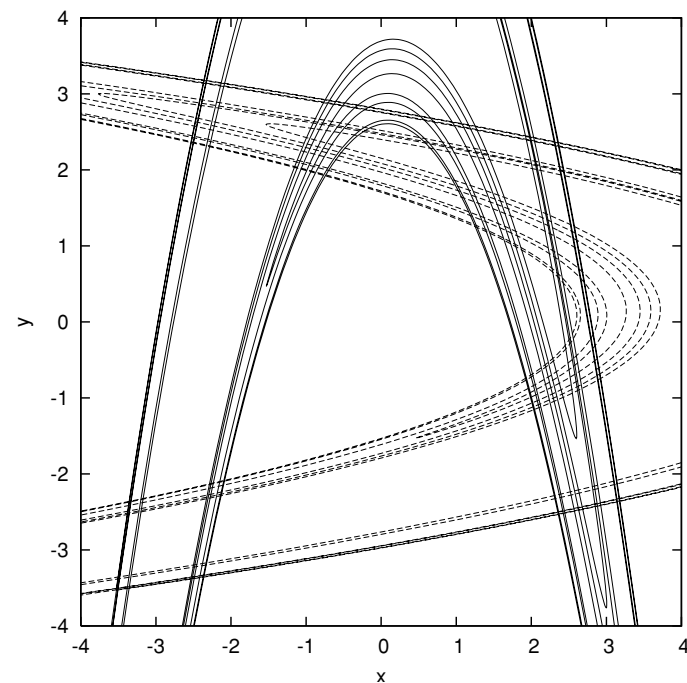
Phase Space



"Symbolic Plane"



Complete Horseshoe
There is a 1 to 1 correspondence



Incomplete Horseshoe?
Some Symbols are "pruned" away...

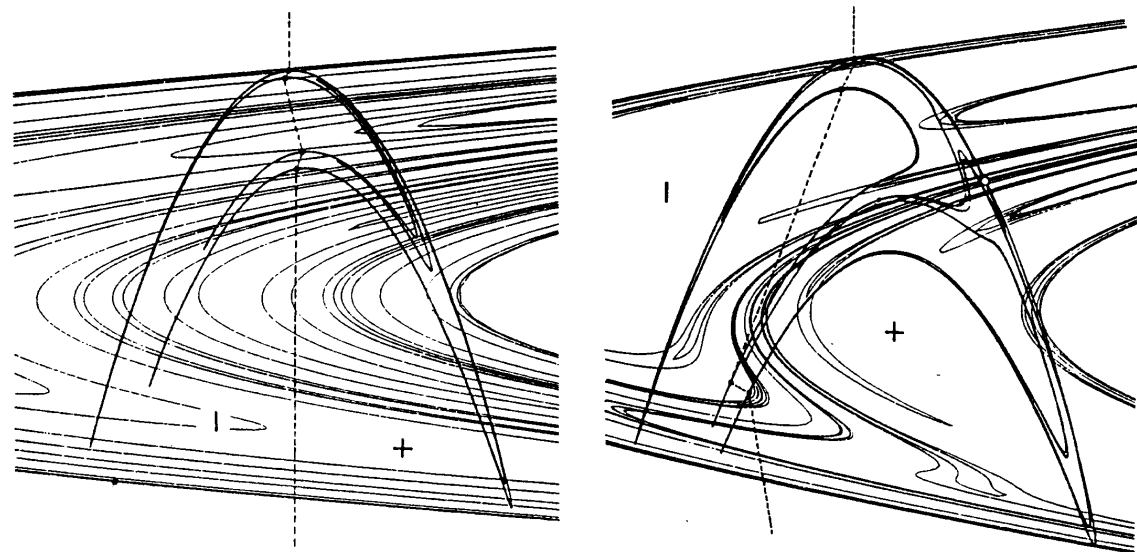
Pruning Front Conjecture



Predrag Cvitanović

Conjecture (P. Cvitanović)

The dynamics of the Henon map can be described as an “incomplete horseshoe” given by a finite pruning front.



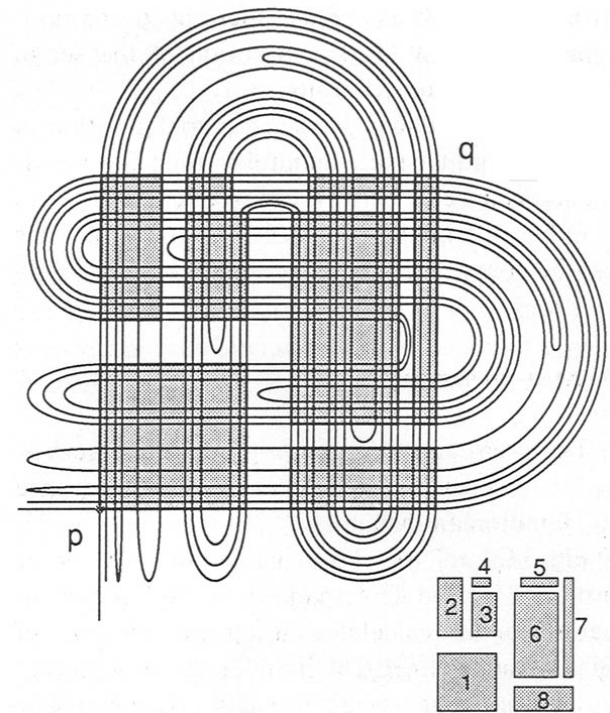
Difficulties...

- * There exist counter examples.
- * No canonical symbolic coding.

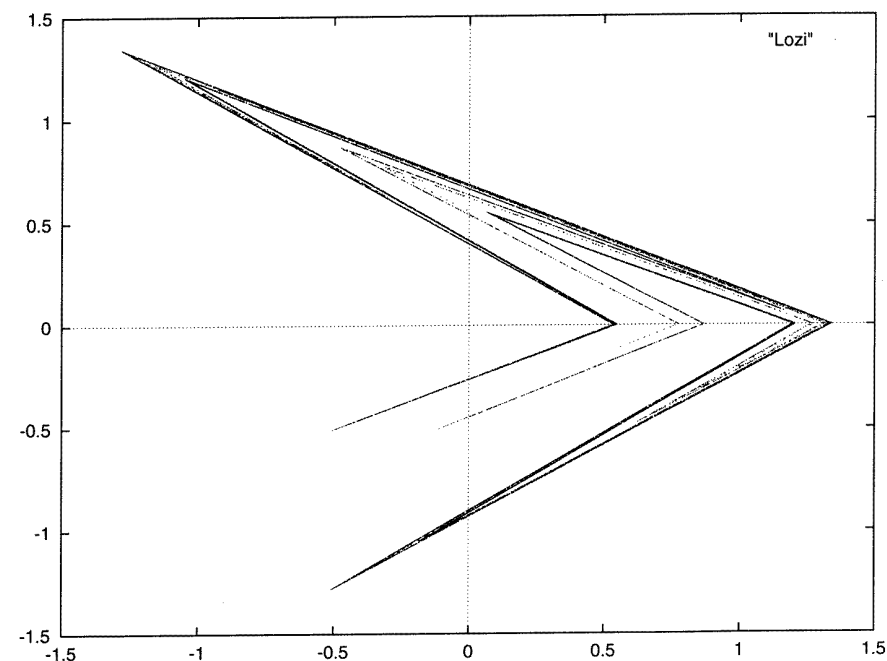
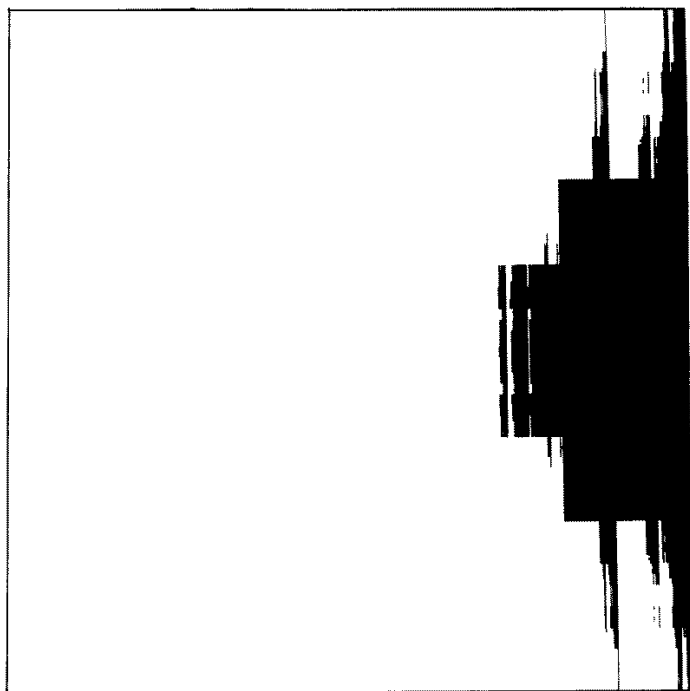
Positive Results

Davis-MacKay-Sannami

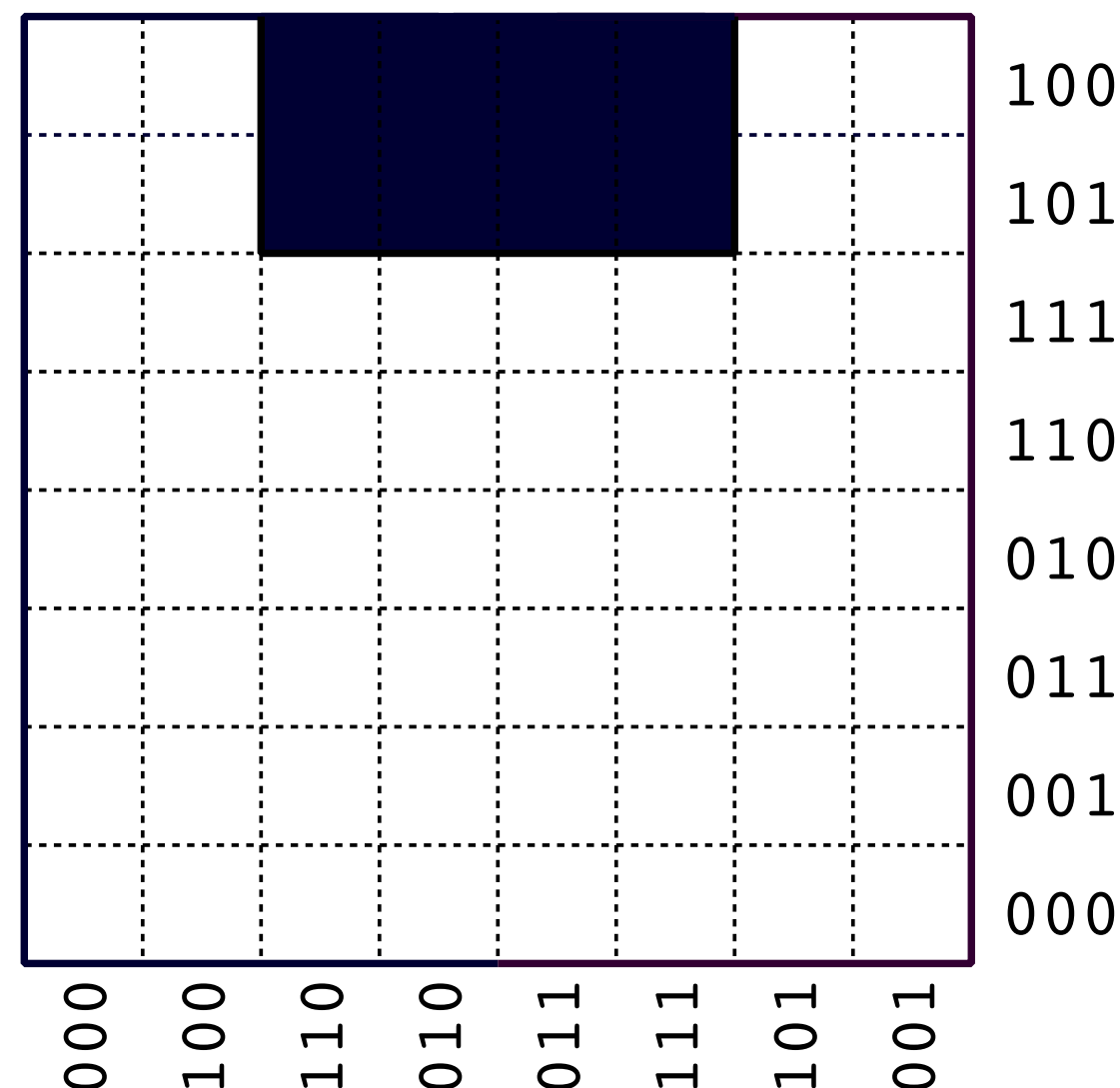
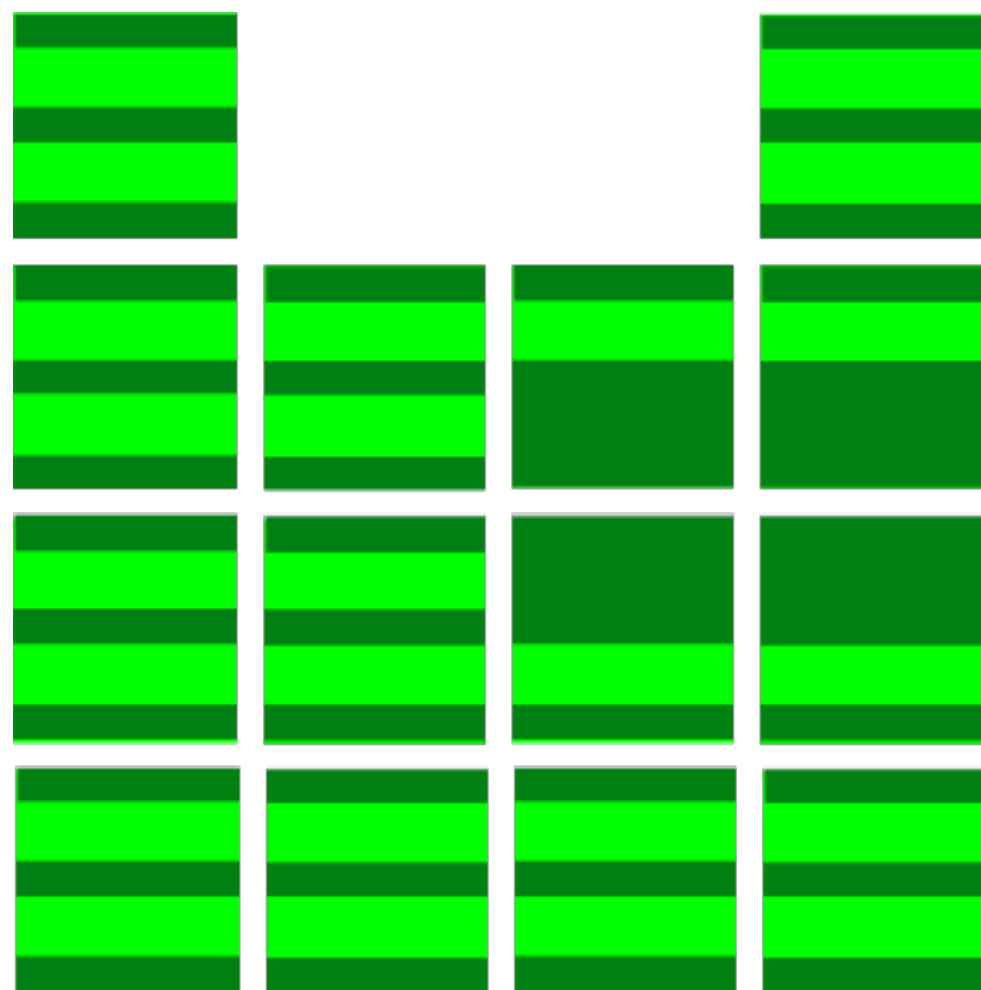
For several parameter values, primary pruned regions and the Markov partitions are constructed and examined.



Yutaka Ishii The conjecture is true for Lozi map.



Index gives an obstruction



The Conley index for this invariant set is given by a 24×24 integer matrix which has 6 non-zero eigenvalues. Hence this set can not be $K_{a,b}^{\mathbb{R}}$ for any $(a, b) \in \mathbb{R}^2$.

Index of the complex Hénon

Theorem. For all $(a, b) \in \mathbb{C}^2$, the Conley index of K is shift equivalent to $2 : \mathbb{Z} \rightarrow \mathbb{Z}$ at $\dim 2$, and trivial at other dimensions.

Sketch of the proof. Choose a real parameter where we know $K^{\mathbb{R}}$ is a hyperbolic horseshoe. Then a complex extension of a real index pair gives a complex index pair. Since a holomorphic map always preserves the orientation, the Conley index is

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} : \mathbb{Z}^2 \rightarrow \mathbb{Z}^2$$

which is shift equivalent to $2 : \mathbb{Z} \rightarrow \mathbb{Z}$.

Sinks and the Julia set as AR decomposition

Attractor Repeller Decomposition

$f : X \rightarrow X$: a homeomorphism

Definition. $A \subset X$ is an attractor if there exists a compact nbd U of A such that $f(U) \subset \text{int } U$.

Definition. For an attractor A , define its dual repeller by $A^* := \{x \in X \mid \omega(x) \cap A = \emptyset\}$. The pair (A, A^*) is called an attractor–repeller decomposition.

Here $\omega(x)$ is defined by

$$\omega(x) = \bigcap_{N>0} \text{cl} \left(\bigcup_{n>N} \{f^n(x)\} \right)$$

Long Exact Sequence

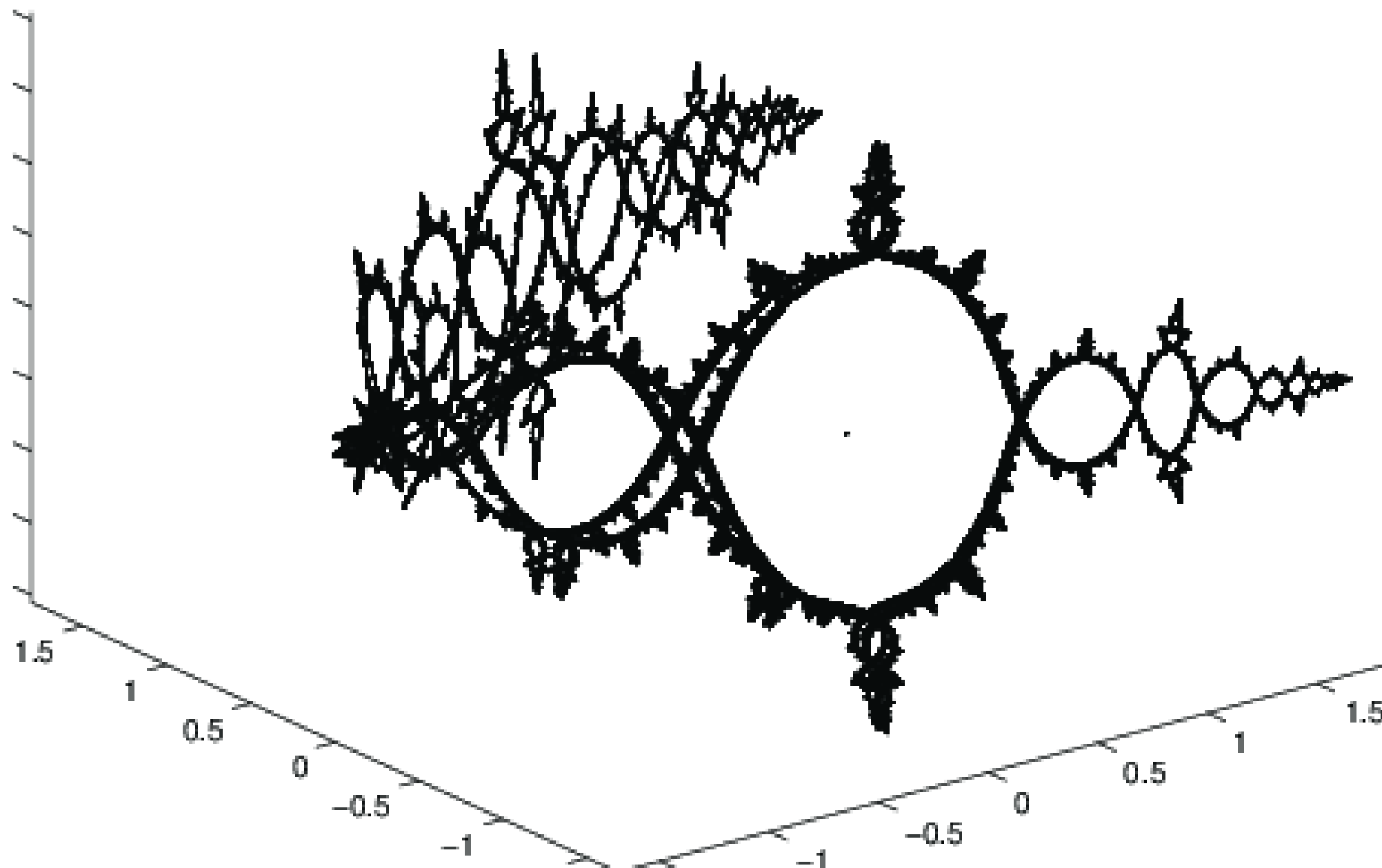
Theorem. Let (A, A^*) be an attractor-repeller decomposition of an isolated invariant set S . Then there exists a long exact sequence

$$\cdots \xrightarrow{\delta_*} \text{Con}_k(A) \rightarrow \text{Con}_k(S) \rightarrow \text{Con}_k(R) \xrightarrow{\delta_*} \text{Con}_{k-1}(A) \rightarrow \cdots$$

where $\text{Con}_*(S)$ is the pair of a vector space and a linear automorphism on it obtained by restricting the pair $f_{P*} : H_*(N/L, [L]; \mathbb{F}) \rightarrow H_*(N/L, [L]; \mathbb{F})$ to its Jordan form away from 0 ($\mathbb{F}$ is a field).

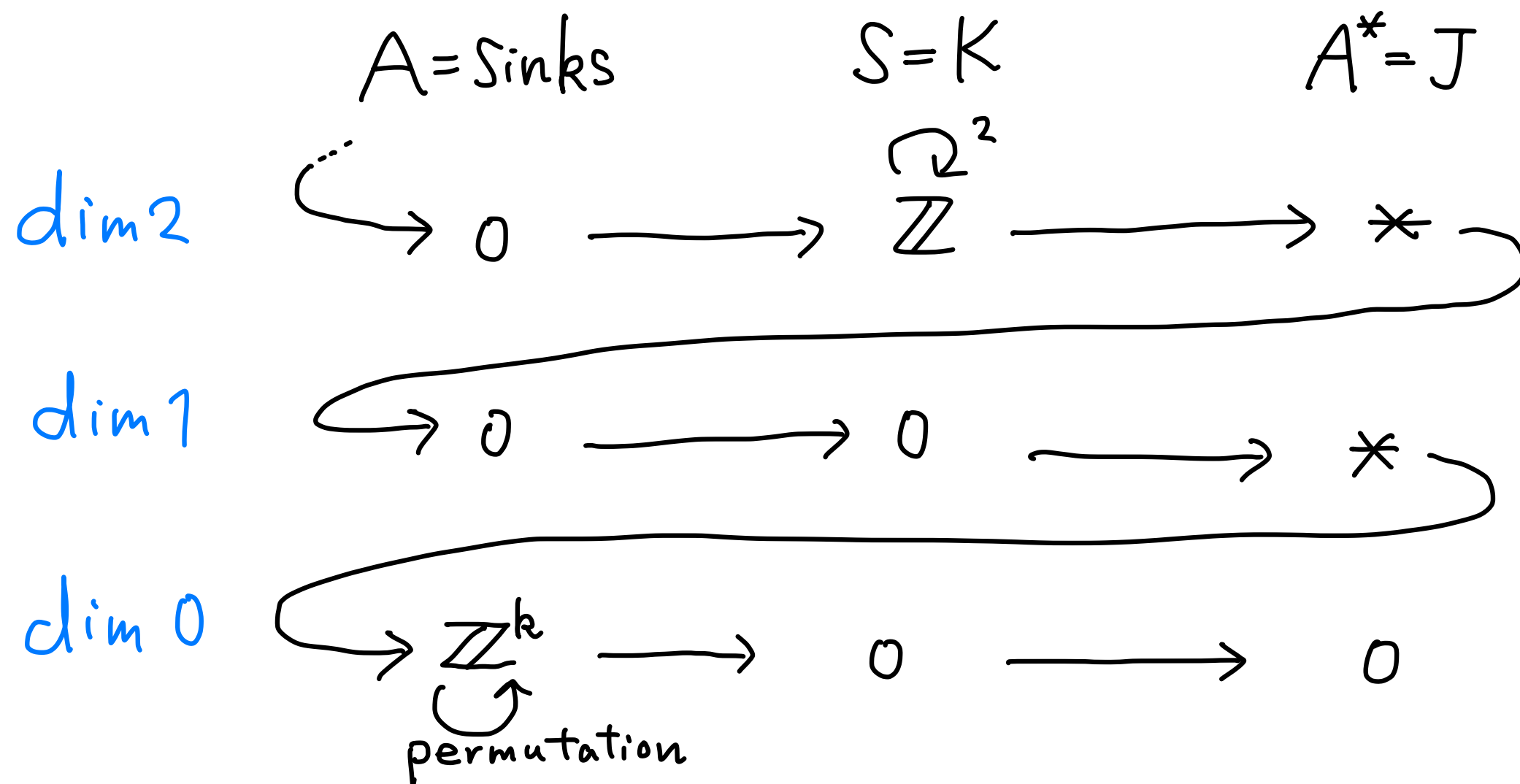
Result

Theorem. Assume that the Hénon map has some attracting periodic orbits and is hyperbolic on J . Then the topological dimension of J is 1.



Proof

- When the map is hyperbolic, the dual repeller of sinks coincides with J (Bedford-Smillie 1991)
- The Conley index of J is non-trivial at dim 1 and 2.



Proof

- ▶ The Conley index of a 0-dim hyperbolic basic set with unstable dim. u should be trivial except for dim. u (de Rezende-Villapouca 2017, Bowen-Franks 1977)
- ▶ This implies that the top. dim. of J should be ≥ 1
- ▶ However, it is known that the Hausdorff dim of a hyperbolic Julia set is < 2 (Verjovsky-Wu 1993).
- ▶ It follows that the topological dim. of J should be 1.

Summary and Future Work

Summary. The Conley index can capture topological restriction on the shape of real and complex Julia sets.

Future Work. A real dynamical system embedded in its complex extension can be understood as a fixed point of the $\mathbb{Z}_2$ action induced by the complex conjugation. We then want to develop a theory of an “equivariant” version of the Conley index and, by using it, generalize the argument today for general group actions.